Worksheet & Answer Key

The enrichment of ores is a process that involves increasing the concentration of valuable metals like _____ and _____ in low-grade ores. This is done to make the extraction of these metals more economical. The process typically involves physical separation techniques, such as flotation, magnetic separation, and electrostatic separation, as well as chemical treatment methods. The enriched ore can then be smelted or refined to produce a higher purity metal.

Word Bank:

manganese, copper

Fill in the blanks:

1. The enrichment of ores is a process that involves increasing the concentration of valuable metals like _____ and copper in low-grade ores.

2. The process typically involves physical separation techniques, such as flotation, magnetic separation, and electrostatic separation, as well as chemical treatment methods to enrich the _____ ore.

3. The enriched ore can then be smelted or refined to produce a higher purity _____ metal.

4. Physical separation techniques, such as flotation, are used to separate valuable metals from the waste rock in low-grade _____ ores.

5. Chemical treatment methods are used to extract valuable metals from low-grade ores by reacting them with a strong _____ solution.

6. The enrichment of ores is necessary to make the extraction of valuable metals like manganese and copper more _____.

7. The enriched ore can then be smelted or refined to produce a higher purity metal, such as _____ or _____

8. Low-grade ores may contain small amounts of valuable metals that are not economically viable for extraction without enrichment techniques like _____.

9. Magnetic separation is used to separate iron and steel waste from low-grade ores containing valuable metals like _____.

■ Answer Key (for teachers only):

1. manganese
2. copper
3. manganese
4. strong
5. economical
6. copper
7. manganese
8. magnetic